

layout of the machines. The manufacturing process is geared towards maximum productivity and not determined by the constraints of rigid automation.”

Some commercially-available robots

Wilson mentions, “Sastra provides solutions to automate the testing of real devices. Traditional robots are used for manufacturing processes. We employ robots to test devices like smartphones, car infotainment systems, consumer electronics like printers and copying machines, and even avionics control panels. These devices are tested for reliability as well as functionality. Testing when done by humans is time consuming and prone to errors. Our robots test these devices in a much faster and reliable way, and generate test reports based on test results.”

Maheswari adds, “A home assistance robot is the next-generation smart, interactive and social family robot. It can play with kids, manage schedules, help with household work, and keep the home safe and connected. It can aid in tasks such as cooking and yoga, using an interactive projector. The multimedia projector can convert any wall in to a movie screen. It also lets you play interactive games.

“The robot converts your home into a smart one, enabling you to control compatible lights, doors and appliances. It comes with a storyteller feature that can read stories. The robot patrols your home using powerful sen-



Kuka innovation for automotive industry

sors and smart intruder alerts.”

Some latest household assistant robots are KURI, Aeolus, Ubtech Robotics Walker, Buddy by Blue Frog Robotics, Sony’s Aibo, Sophia by Hanson Robotics, Aido and Piaggio.

Shoran adds, “Kuka’s automation is the decisive key to higher productivity, higher reliability and greater cost-effectiveness.

“Robots have been extensively used in sectors such as food, plastics, metal, electronics, medical and entertainment, besides the automotive industry. These can perform transfer tasks, even in small batch sizes, safely and without errors using information received from the cloud. These can then report the experiences back to the cloud, where these can be evaluated for optimisation, documentation and quality-assurance purposes. Not only the robots, but also machine tools and other devices in the manufacturing facility can be connected to Kuka cloud to collect and analyse data for optimising the production processes.”

Examples of such robots include the ones by Infineon, Bajaj Auto, Aurolab, Sastra and ABB.

Machine learning

Machine learning has had a significant impact on smart robotic technologies. It integrates vision, imitation learning, self-supervised learning, and assistive and medical technologies.

Robot vision helps in structured predic-

tion learning techniques and inspection systems like identification and sorting of objects.

Imitation learning provides observational learning used for humanoid robots. It is required for construction, agriculture, search and rescue, military and other areas.

Self-supervised learning enables robots to generate their own training courses to improve performance. It includes road detection algo-

rithms and more to improve performance.

Machine learning-based assistive technologies are being developed to combine assistive machines with autonomy, to help people with disabilities and the elderly.

Robotic applications are enhanced when barcode readers and machine vision are added to these. A smart camera on a robot is required for machine learning. It acts as a set of eyes that help the robot compare the image of a product against a set of criteria, and make sure that the criteria are met.

Mounting a camera on an industrial arm provides a variety of features for inspection. Robotic applications that require a smart camera are pick-and-place, assembly, packaging and palletising, quality inspection, screw driving, polishing, labelling, welding, molding, lab testing and others. For example, in assembly, machine vision guides the parts to be assembled, while the barcode reader reads the label on the parts to make sure correct parts are assembled.

Conclusion

We are living in a transformative and an automotive era, where advancements in technology are leading to wider adoption of smart robotics. We are constantly in touch with new forms of technology that make daily life easier. Robotic automation is replacing heavy-duty and risky tasks performed by humans. Automation will lead to many industrial jobs overtaken by robots. At the same time, new and advanced technology jobs will be created elsewhere. **EFY**



LBR iiwa, a robotic assistant, working collaboratively with humans to increase productivity in manufacturing and materials handling